FIITJEE JEE MAINS MOCK TEST-1

Target IIT - JEE 2014
QP CODE: 120634.0

Time : 3 hours ***Maximum Marks : 360***

- ***Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.***
- ***You are not allowed to leave the examination hall before end of the test.***

Instructions

1. Immediately fill in the particulars on this page of the Test Booklet with *Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.*

2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.

3. The test is of **3 hours** duration.

4. The Test Booklet consists of **90** questions. The maximum marks are **360**.

5. There are ***three*** parts in the question paper A, B, C consisting of **Physics, Chemistry** and **Mathematics** having 30 questions in each part of equal weightage. Each question is allotted **4 (four)** marks for correct response.

6. *Candidates will be awarded marks as stated above in instruction No.5 for correct response of each question. ¼ (one fourth) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.*

7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instruction 6 above.

8. Use ***Blue / Black Ball Point Pen only*** for writing particulars / marking responses on ***Side-1*** and ***Side-2*** of the Answer Sheet. ***Use of pencil is strictly prohibited.***

9. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. except the Admit Card inside the examination hall / room.

10. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room / Hall. ***However, the candidates are allowed to take away this Test Booklet with them.***

11. **Do not fold or make any stray marks on the Answer Sheet.**

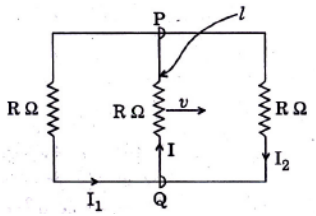
Name of the Candidate (in Capital Letters) :_____

Enrolment Number :_____

Batch :_____ **Date of Examination :** _____

SECTION - 1
PHYSICS

1. A rectangular loop has a sliding connector PQ of length ℓ and resistance RΩ and it is moving with a speed v as shown. The set-up is placed in a uniform magnetic field going into the plane of the paper. The three currents I_1, I_2 and I are



(A) $I_1 = -I_2 = \frac{B\ell v}{R}, I = \frac{2B\ell v}{R}$

(B) $I_1 = I_2 = \frac{B\ell v}{3R}, I = \frac{2B\ell v}{3R}$

(C) $I_1 = I_2 = I = \frac{B\ell v}{R}$

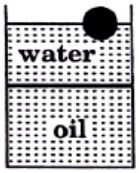
(D) $I_1 = I_2 = \frac{B\ell v}{6R}, I = \frac{B\ell v}{3R}$

2. Let C be the capacitance of a capacitor discharging through a resistor R. Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio t_1/t_2 will be

(A) 1 (B) $\frac{1}{2}$

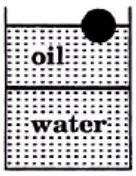
(C) $\frac{1}{4}$ (D) 2

3. A ball is made of a material of density ρ where $\rho_{oil} < \rho < \rho_{water}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?

(A) 

(B) 

(C) 

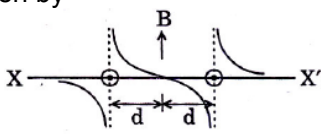
(D) 

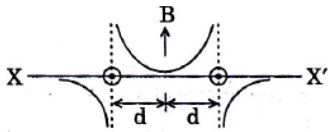
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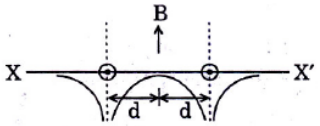
4. A particle is moving with velocity $\vec{v} = K(y\hat{i} + x\hat{j})$, where K is a constant. The general equal for its path is

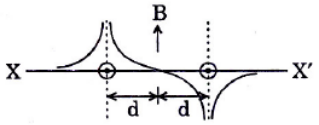
(A) $y = x^2$ + constant
(B) $y^2 = x$ + constant
(C) xy = constant
(D) $y^2 = x^2$ + constant

5. Two long parallel wires are at a distance 2d apart. They carry steady equal current flowing out of the plane of the paper as shown. The variation of the magnetic field along the line XX′ is given by

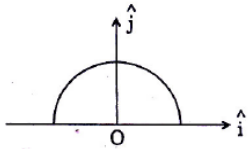
(A)


(B)


(C)


(D)


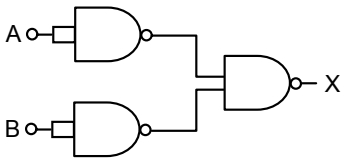
6. A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is



(A) $\frac{q}{4\pi^2\varepsilon_o r^2}\hat{j}$
(B) $-\frac{q}{4\pi^2\varepsilon_o r^2}\hat{j}$
(C) $-\frac{q}{2\pi^2\varepsilon_o r^2}\hat{j}$
(D) $\frac{q}{2\pi^2\varepsilon_o r^2}\hat{j}$

7. The combination of gates shown below yields

(A) OR gate
(B) NOT gate
(C) XOR gate
(D) NAND gate



8. A diatomic ideal gas is used in a car engine as the working substance. If during the adiabatic expansion part of the cycle, volume of the gas increases from V to 32V the efficiency of the engine is

(A) 0.5
(B) 0.75
(C) 0.99
(D) 0.25

Space for rough work

9. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_o\left(\frac{5}{4} - \frac{r}{R}\right)$ upto r = R and $\rho(r)$ = 0 for r > R, where r is the distance from the origin. The electric field at a distance r(r < R) from the origin is given by

(A) $\frac{4\pi\rho_o r}{3\varepsilon_o}\left(\frac{5}{3} - \frac{r}{R}\right)$ (B) $\frac{\rho_o r}{4\varepsilon_o}\left(\frac{5}{3} - \frac{r}{R}\right)$

(C) $\frac{4\rho_o r}{3\varepsilon_o}\left(\frac{5}{4} - \frac{r}{R}\right)$ (D) $\frac{\rho_o r}{3\varepsilon_o}\left(\frac{5}{4} - \frac{r}{R}\right)$

10. The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are constants and x is the distance between the atoms. If the dissociation energy of the molecule is $D = [U(x = \infty) - U_{\text{at equilibrium}}]$, D is

(A) $\frac{b^2}{2a}$ (B) $\frac{b^2}{12a}$

(C) $\frac{b^2}{4a}$ (D) $\frac{b^2}{6a}$

11. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8 g cm^{-3}, the angle remains the same. If density of the material of the sphere is 16 g cm^{-3}, the dielectric constant of the liquid is

(A) 4 (B) 3

(C) 2 (D) 1

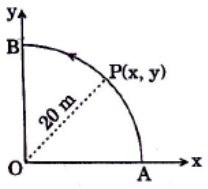
12. Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2. The respective temperature coefficients of their series and parallel combinations are nearly

(A) $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$ (B) $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$

(C) $\alpha_1 + \alpha_2, \frac{\alpha_1\alpha_2}{\alpha_1 + \alpha_2}$ (D) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$

Space for rough work

13. A point P moves in counter-clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a lengths $s = t^3 + 5$, where s is in metres an t is in seconds. The radius of the path is 20 m. The acceleration of 'P' when t = 2s is nearly



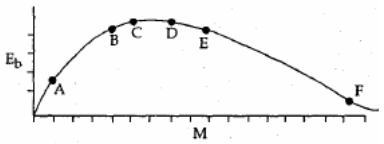
(A) 13 m/s^2 (B) 12 m/s^2
(C) 7.2 m/s^2 (D) 14 m/s^2

14. For a particle in uniform circular motion the acceleration $\vec{a}$ at a point P(R, θ) on the circle of radius R is (here θ is measured from the x-axis)

(A) $-\frac{v^2}{R}\cos\theta\hat{i} + \frac{v^2}{R}\sin\theta\hat{j}$ (B) $-\frac{v^2}{R}\sin\theta\hat{i} + \frac{v^2}{R}\cos\theta\hat{j}$

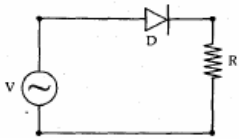
(C) $-\frac{v^2}{R}\cos\theta\hat{i} - \frac{v^2}{R}\sin\theta\hat{j}$ (D) $\frac{v^2}{R}\hat{i} + \frac{v^2}{R}\hat{j}$

15. The above is a plot of binding energy per nucleon E_b, against the nuclear mass M; A, B,, C, D,, E, F correspond to different nuclei. Consider four reactions:



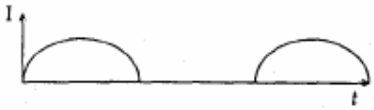
(i) $A + B \rightarrow C + \varepsilon$ (ii) $C \rightarrow A + B + \varepsilon$
(iii) $D + E + \varepsilon$ and (iv) $F \rightarrow D + E + \varepsilon$
where ε is the energy released? In which reactions is ε positive?
(A) (i) and (iv) (B) (i) and (iii)
(C) (ii) and (iv) (D) (ii) and (iii)

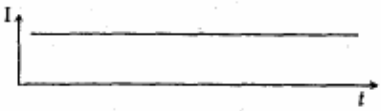
16. A p-n junction (D) shown in the figure can act as a rectifier. An alternating current source (V) is connected in the circuit.



(A) 

(B) 

(C) 

(D) 

Space for rough work

17. If x, v and a denote the displacement, the velocity and the acceleration of a particle executing simple harmonic motion of time period T, then, which of the following does not change with time?

(A) $a^2T^2 + 4\pi^2v^2$ (B) $\frac{aT}{x}$

(C) $aT + 2\pi v$ (D) $\frac{aT}{v}$

18. In an optics experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v, from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x-axis meets the experimental curve at P. The coordinates of P will be

(A) (2f, 2f) (B) $\left(\frac{f}{2}, \frac{f}{2}\right)$

(C) (f, f) (D) (4f, 4f)

19. A thin uniform rod of length ℓ and mass m is swinging freely about a horizontal axis passing through its end. Its maximum angular speed is ω. Its centre of mass rises to a maximum height of

(A) $\frac{1}{3}\frac{\ell^2\omega^2}{g}$ (B) $\frac{1}{6}\frac{\ell\omega}{g}$

(C) $\frac{1}{2}\frac{\ell^2\omega^2}{g}$ (D) $\frac{1}{6}\frac{\ell^2\omega^2}{g}$

20. Let $P(r) = \frac{Q}{\pi R^4}r$ be the charge density distribution for a solid sphere of radius R and total charge Q for a point 'p' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is

(A) 0 (B) $\frac{Q}{4\pi\varepsilon_o r_1^2}$

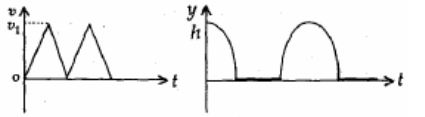
(C) $\frac{Qr_1^2}{4\pi\varepsilon_o R^4}$ (D) $\frac{Qr_1^2}{3\pi\varepsilon_o R^4}$

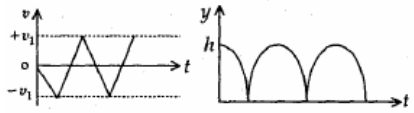
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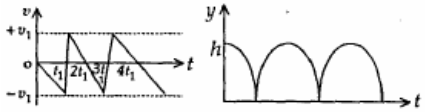
21. One kg of a diatomic gas is at a pressure of 8×10^4 N/m^2. The density of the gas is 4 kg/m^{-3}. What is the energy of the gas due to its thermal motion?
(A) 3×10^4 J (B) 5×10^4 J
(C) 6×10^4 J (D) 7×10^4 J

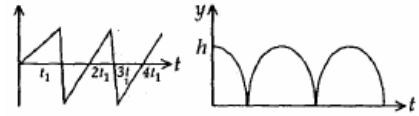
22. A motor cycle starts from rest and accelerates along a straight path at 2 m/s^2. At the starting point of the motor cycle there is a stationary electric sire. How far has the motor cycle gone when the driver hears the frequency of the siren at 94% of its value when the motor cycle was at rest? (speed of sound = 330 ms^{-1}).
(A) 49 m (B) 98 m
(C) 147 m (D) 196 m

23. Consider a rubber ball freely falling from a height h = 4.9 m onto a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic. Then the velocity as a function of time the height as function of time will be

(A) 

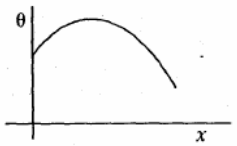
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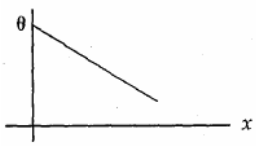
(C) 

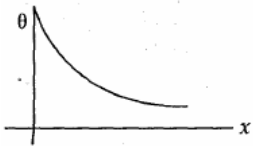
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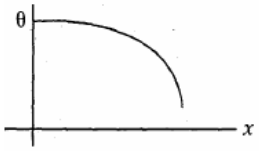
24. A charge Q is placed at each of the opposite corners of a square. A charge q is placed at each of the other two corners. If the net electrical force on Q is zero, then the Q/q equals
(A) $-2\sqrt{2}$ (B) –1
(C) 1 (D) $-\frac{1}{\sqrt{2}}$

25. A long metallic bar is carrying heat from one of its ends to the other end under steady-state. The variation of temperature θ along the length x of the bar from its hot end is best described by which of the following figure.

(A) 

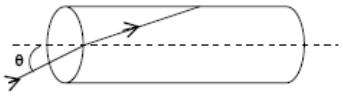
(B) 

(C) 

(D) 

Space for rough work

26. A transparent solid cylindrical rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid point of one end of the rod as shown in the figure. The incident angle θ for which the light ray grazes along the wall of the rod is



(A) $\sin^{-1}\left(\frac{1}{2}\right)$ (B) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(C) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

27. Three sound waves of equal amplitudes have frequencies (v – 1), v, (v + 1). They superpose to give beats. The number of beats produced per second will be

(A) 4 (B) 3

(C) 2 (D) 1

28. Two wires are made of the same material and have the same volume. However wire 1 has crossectional area A and wire-2 has cross-sectional area 3A. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount?

(A) F (B) 4F

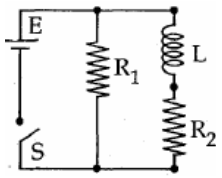
(C) 6F (D) 9F

29. In an experiment the angles are required to be measured using an instrument. 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half-a-degree(=0.5o), then the least count of the instrument is

(A) one minute (B) half minute

(C) one degree (D) half degree

30. An inductor of inductance L = 400 mH and resistors of resistances $R_1 = 2\Omega$ and $R_2 = 2\Omega$ are connected to a battery of emf 12V as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at t = 0. The potential drop across L as a function of time is



(A) $6e^{-5t}$ V (B) $\frac{12}{t}e^{-3t}$ V

(C) $6(1 - e^{-1/02})$ V (D) $12e^{-5t}$ V

Space for rough work

SECTION - 2
CHEMISTRY

1. Knowing that the Chemistry of lanthanoids (Ln) is dominated by its +3 oxidation state, which of the following statement is incorrect ?
   (A) Because of the large size of the Ln (III) ions the bonding in its compounds is predominantly ionic in character.
   (B) The ionic sizes of Ln (III) decrease in general with increasing atomic number.
   (C) Ln (III) compounds are generally colourless.
   (D) Ln (III) hydroxides are mainly basic in character.

2. A liquid was mixed with ethanol and a drop of concentrated H_2SO_4 was added. A compound with a fruity smell was formed. The liquid was :
   (A) CH_3OH (B) HCHO
   (C) CH_3COCH_3 (D) CH_3COOH

3. In which of the following arrangements, the sequence is not strictly according to the property written against it?
   (A) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power
   (B) HF < HCl < HBr < HI: increasing acid strength
   (C) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength
   (D) B < C < O < N: increasing first ionization enthalpy.

4. Which of the following statements is incorrect regarding physiosorption ?
   (A) It occurs because of van der Waal's forces.
   (B) More easily liquefiable gases are adsorbed readily.
   (C) Under high pressure it results into multi molecular layer on adsorbent surface.
   (D) Enthalpy of adsorption ($\Delta H_{adsorption}$) is low and positive.

5. Given: $E^{o}_{Fe^{3+}/Fe} = -0.036\,V$, $E^{o}_{Fe^{2+}/Fe} = -0.439\,V$. The value of standard electrode potential for the change, $Fe^{3+}(aq) + e^{-} \longrightarrow Fe^{2+}(aq)$ will be :
   (A) -0.072 V (B) 0.385 V
   (C) 0.770 V (D) -0.270 V

6. Solid $Ba(NO_3)_2$ is gradually dissolved in a 1.0×10^{-4} M Na_2CO_3 solution. At what concentration of Ba^{2+} will a precipitate begin to form? (K_{sp} for $BaCO_3 = 5.1 \times 10^{-9}$).
   (A) 4.1×10^{-5} M (B) 5.1×10^{-5} M
   (C) 8.1×10^{-8} M (D) 8.1×10^{-7} M

Space for rough work

7. Which one of the following reactions of Xenon compounds is not feasible ?
(A) $XeO_3 + 6HF \longrightarrow XeF_6 + 3H_2O$
(B) $3XeF_4 + 6H_2O \longrightarrow 2Xe + XeO_3 + 12HF + 1.5O_2$
(C) $2XeF_2 + 2H_2O \longrightarrow 2Xe + 4HF + O_2$
(D) $XeF_6 + RbF \longrightarrow Rb[XeF_7]$

8. The set representing the correct order of ionic radius is :
(A) $Li^+ > Be^{2+} > Na^+ > Mg^{2+}$ (B) $Na^+ > Li^+ > Mg^{2+} > Be^{2+}$
(C) $Li^+ > Na^+ > Mg^{2+} > Be^{2+}$ (D) $Mg^{2+} > Be^{2+} > Li^+ > Na^+$

9. The bond dissociation energy of B – F in BF_3 is 646 kJ mol^{-1} whereas that of C-F in CF_4 is 515 kJ mol^{-1}. The correct reason for higher B-F bond dissociation energy as compared to that of C- F is :
(A) smaller size of B-atom as compared to that of C- atom
(B) stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4
(C) significant $p\pi$ - $p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4 .
(D) lower degree of $p\pi$ - $p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4 .

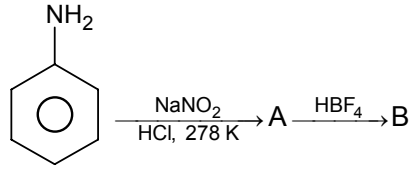
10. Which of the following pairs represents linkage isomers ?
(A) $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4]$ $[CuCl_4]$
(B) $[Pd(PPh_3)_2 (NCS)_2]$ and $[Pd(PPh_3)_2 (SCN)_2]$
(C) $[Co(NH_3)_5NO_3]SO_4$ and $[Co(NH_3)_5SO_4]NO_3$
(D) $[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$

11. If 10^{-4} dm^3 of water is introduced into a 1.0 dm^3 flask at 300 K, how many moles of water are in the vapour phase when equilibrium is established ?
(Given : Vapour pressure of H_2O at 300 K is 3170 Pa; R = 8.314 J K^{-1} mol^{-1})
(A) 5.56×10^{-3} mol (B) 1.53×10^{-2} mol
(C) 4.46×10^{-2} mol (D) 1.27×10^{-3} mol

12. From amongst the following alcohols the one that would react fastest with conc. HCl and anhydrous $ZnCl_2$, is
(A) 2–Butanol (B) 2–Methylpropan–2–ol
(C) 2–Methylpropanol (D) 1–Butanol

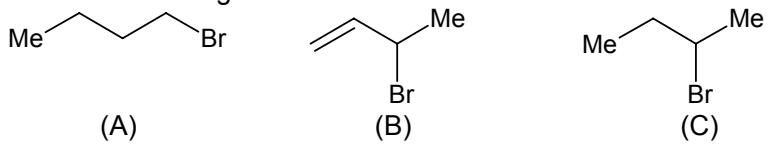
Space for rough work

13. In the chemical reactions,



the compounds 'A' and 'B' respectively are
(A) nitrobenzene and fluorobenzene
(B) phenol and benzene
(C) benzene diazonium chloride and fluorobenzene
(D) nitrobenzene and chlorobenzene

14. Consider the following bromides:

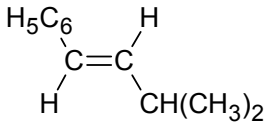


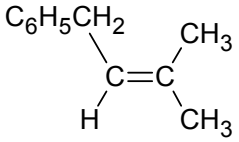
(A) (B) (C)

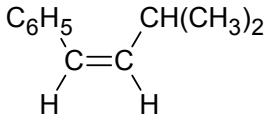
The correct order of S_N1 reactivity is
(A) B > C > A
(B) B > A > C
(C) C > B > A
(D) A > B > C

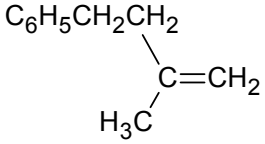
15. The main product of the following reaction is

$$C_6H_5CH_2CH(OH)CH(CH_3)_2 \xrightarrow{Conc.H_2SO_4} ?$$

(A) 

(B) 

(C) 

(D) 

16. The presence or absence of hydroxyl group on which carbon atom of sugar differentiates RNA and DNA?
(A) 2nd
(B) 3rd
(C) 4th
(D) 1st

Space for rough work

17. Trichloroacetaldehyde was subjected to Cannizzaro's reaction by using NaOH. The mixture of the products contains sodium trichloroacetate and another compound. The other compound is :
(A) trichloromethanol
(B) 2, 2, 2-trichloropropanol
(C) chloroform
(D) 2, 2, 2-trichloroethanol

18. 'a' and 'b' are van der Waals' constants for gases. Chlorine is more easily liquefied than ethane because
(A) a and b for Cl_2 < a and b for C_2H_6
(B) a for Cl_2 < a for C_2H_6 but b for Cl_2 > b for C_2H_6
(C) a for Cl_2 > a for C_2H_6 but b for Cl_2 < b for C_2H_6
(D) a and b for Cl_2 > a and b for C_2H_6

19. In context of the lanthanoids, which of the following statement is not correct ?
(A) All the members exhibit +3 oxidation state
(B) Because of similar properties the separation of lanthanoids is not easy.
(C) Availability of 4f electrons results in the formation of compounds in +4 state for all the members of the series.
(D) There is a gradual decrease in the radii of the members with increasing atomic number in the series.

20. Which of the following statements regarding sulphur is incorrect ?
(A) The vapour at 200°C consists mostly of S_8 rings
(B) At 600°C the gas mainly consists of S_2 molecules
(C) The oxidation state of sulphur is never less than +4 in its compounds
(D) S_2 molecule is paramagnetic.

21. Phenol is heated with a solution of mixture of KBr and $KBrO_3$. The major product obtained in the above reaction is
(A) 3-Bromophenol
(B) 4-Bromophenol
(C) 2, 4, 6- Tribromophenol
(D) 2-Bromophenol

22. The molecule having smallest bond angle is :
(A) NCl_3
(B) $AsCl_3$
(C) $SbCl_3$
(D) PCl_3

23. The incorrect expression among the following is:
(A) $\frac{\Delta G_{system}}{\Delta S_{total}} = -T$
(B) In isothermal process $W_{reversible} = -nRT\ln\frac{V_2}{V_1}$
(C) $\ln K = \frac{\Delta H^0 - T\Delta S^0}{RT}$
(D) $K = e^{-\Delta G^0/RT}$

Space for rough work

24. The density of a solution prepared by dissolving 120 g of urea (mol. Mass = 60 u) in 1000g of water is 1.15 g/mL. The molarity of this solution is:
(A) 0.50 M (B) 1.78 M
(C) 1.02 M (D) 2.05 M

25. How many chiral compounds are possible on monochlorination of 2–methyl butane ?
(A) 8 (B) 2
(C) 4 (D) 6

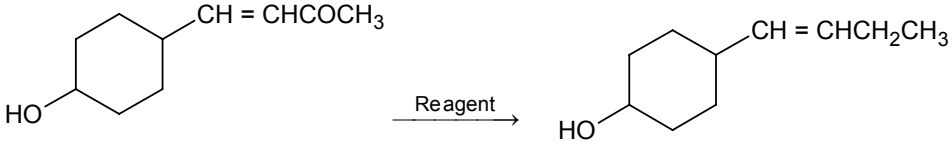
26. For a first order reaction, (A) $\rightarrow$ products, the concentration of A changes from 0.1 M to 0.025 M in 40 minutes. The rate of reaction when the concentration of A is 0.01 M is :
(A) 1.73×10^{-5} M/ min (B) 3.47×10^{-4} M/min
(C) 3.47×10^{-5} M/min (D) 1.73×10^{-4} M/min

27. Iron exhibits + 2 and +3 oxidation states. Which of the following statements about iron is incorrect ?
(A) Ferrous oxide is more basic in nature than the ferric oxide.
(B) Ferrous compounds are relatively more ionic than the corresponding ferric compounds
(C) Ferrous compounds are less volatile than the corresponding ferric compounds.
(D) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.

28. K_f for water is 1.86 K kg mol^{-1}. If your automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol ($C_2H_6O_2$) must you add to get the freezing point of the solution lowered to –2.8°C ?
(A) 72g (B) 93g
(C) 39g (D) 27g

29. Iodoform can be prepared from all except :
(A) Ethyl methyl ketone (B) Isopropyl alcohol
(C) 3–Methyl – 2– butanone (D) Isobutyl alcohol

30. In the given transformation, which of the following is the most appropriate reagent?



(A) NH_2NH_2, OH^- (B) Zn-Hg/HCl
(C) Na.Liq.NH_3 (D) $NaBH_4$

Space for rough work

SECTION - 3
MATHEMATICS

1. The real number k for which the equation, $2x^3 + 3x + k = 0$ has two distinct real roots in [0, 1]
   (A) lies between –1 and 0 (B) does not exist
   (C) lies between 1 and 2 (D) lies between 2 and 3

2. If $A = \sin^2 x + \cos^4 x$, then for all real x
   (A) $\frac{13}{16} \le A \le 1$ (B) $1 \le A \le 2$
   (C) $\frac{3}{4} \le A \le \frac{13}{16}$ (D) $\frac{3}{4} \le A \le 1$

3. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to:
   (A) 5 (B) 0
   (C) 4 (D) 11

4. $\lim\limits_{x \to 2} \left(\frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2} \right)$
   (A) equals $\sqrt{2}$ (B) equals $-\sqrt{2}$
   (C) equals $\frac{1}{\sqrt{2}}$ (D) does not exist

5. At present a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is:
   (A) 3500 (B) 4500
   (C) 2500 (D) 3000

6. ABCD is a trapezium such that AB and CD are parallel and $BC \perp CD$. If $\angle ADB = \theta$, BC = p and CD = q, then AB is equal to
   (A) $\frac{p^2 + q^2}{p^2 \cos\theta + q^2 \sin\theta}$ (B) $\frac{(p^2 + q^2)\sin\theta}{(p\cos\theta + q\sin\theta)^2}$
   (C) $\frac{(p^2 + q^2)\sin\theta}{p\cos\theta + q\sin\theta}$ (D) $\frac{p^2 + q^2\cos\theta}{p\cos\theta + q\sin\theta}$

Space for rough work

7. The value of $\int_0^1 \frac{8\log(1+x)}{1+x^2}dx$ is

(A) $\frac{\pi}{8}\log 2$ (B) $\frac{\pi}{2}\log 2$

(C) $\log 2$ (D) $\pi\log 2$

8. A ray of light along $x+\sqrt{3}y=\sqrt{3}$ gets reflected upon reaching x-axis, the equation of the reflected ray is:

(A) $y=\sqrt{3}x-\sqrt{3}$ (B) $\sqrt{3}y=x-1$

(C) $y=x+\sqrt{3}$ (D) $\sqrt{3}y=x-\sqrt{3}$

9. The domain of the function $f(x)=\frac{1}{\sqrt{|x|-x}}$ is

(A) $(0,\infty)$ (B) $(-\infty, 0)$

(C) $(-\infty,\infty)-\{0\}$ (D) $(-\infty,\infty)$

10. If z is a complex number of unit modulus and argument θ, then $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ equals:

(A) θ (B) $\pi-\theta$

(C) $-\theta$ (D) $\frac{\pi}{2}-\theta$

11. If $\int f(x)\,dx=\psi(x)$, then $\int x^5 f(x^3)dx$ is equal to:

(A) $\frac{1}{3}x^3\psi(x^3)-\int x^2\psi(x^3)dx+C$ (B) $\frac{1}{3}\left[x^3\psi(x^3)-\int x^3\psi(x^3)dx\right]+C$

(C) $\frac{1}{3}\left[x^3\psi(x^3)-\int x^2\psi(x^3)dx\right]+C$ (D) $\frac{1}{3}x^3\psi(x^3)-3\int x^3\psi(x^3)dx+C$

12. Equation of the ellipse whose axes are the axes of coordinates and which passes through the point (–3, 1) and has eccentricity $\sqrt{\frac{2}{5}}$ is

(A) $5x^2+3y^2-48=0$ (B) $3x^2+5y^2-15=0$

(C) $5x^2+3y^2-32=0$ (D) $3x^2+5y^2-32=0$

Space for rough work

13. If the lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ and $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar, then k can have:
(A) exactly two values (B) exactly three values
(C) any value (D) exactly one value

14. The number of values of k for which the linear equations $4x+ky+2z=0; kx+4y+z=0;$ $2x+2y+z=0$ posses a non – zero solution is
(A) 2 (B) 1
(C) zero (D) 3

15. Consider:
Statement – I : $(p\wedge\sim q)\wedge(\sim p\wedge q)$ is a fallacy.
Statement – II : $(p\to q)\leftrightarrow(\sim q\to\sim p)$ is a tautology.
(A) Statement – I is true; Statement – II is false.
(B) Statement – I is false; Statement – II is true.
(C) Statement – I is true; Statement – II is true; Statement – II is a correct explanation for Statement – I
(D) Statement – I is true; Statement – II is true; Statement – II is not a correct explanation for Statement – I.

16. If the equations $x^2+2x+3=0$ and $ax^2+bx+c=0, a,b,c\in R,$ have a common root, then a : b : c is:
(A) 1 : 3 : 2 (B) 3 : 1 : 2
(C) 1 : 2 : 3 (D) 3 : 2 : 1

17. If the mean deviation about the median of the numbers $\alpha,2\alpha,.....50\alpha$ is 50, then $|\alpha|$ equals
(A) 3 (B) 4
(C) 5 (D) 2

18. The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$ is
(A) 210 (B) 310
(C) 4 (D) 120

19. Let A and B be two symmetric matrices of order 3.
Statement – 1 A (BA) and (AB) A are symmetric matrices.
Statement – 2 AB is symmetric matrix if matrix multiplication of A and B is commutative.
(A) Statement – 1 is true, Statement – 2 is true; Statement – 2 is not a correct explanation for Statement – 1
(B) Statement – 1 is true, Statement – 2 is false
(C) Statement – 1 is false, Statement – 2 is true
(D) Statement – 1 is true, Statement – 2 is true; Statement – 2 is a correct explanation for Statement – 1

Space for rough work

20. If x, y, z are in A.P. and $\tan^{-1}x$, $\tan^{-1}y$ and $\tan^{-1}z$ are also in A.P., then:

(A) $6x = 3y = 2z$ (B) $6x = 4y = 3z$

(C) $x = y = z$ (D) $2x = 3y = 6z$

21. The intercepts on x-axis made by tangents to the curve, $y = \int_0^x |t|dt,\ x \in R$, which are parallel to the line $y = 2x,$ are equal to

(A) ± 3 (B) ± 4

(C) ± 1 (D) ± 2

22. Let [.] denote the greatest integer function then the value of $\int_0^{1.5} x\left[x^2\right]dx$ is:

(A) 0 (B) $\frac{3}{2}$

(C) $\frac{3}{4}$ (D) $\frac{5}{4}$

23. The circle passing through (1, –2) and touching the axis of x at (3, 0) also passes through the point:

(A) $(5, -2)$ (B) $(-2, 5)$

(C) $(-5, 2)$ (D) $(2, -5)$

24. If $A(2, -3)$ and $B(-2, 1)$ are two vertices of a triangle and third vertex moves on the line $2x + 3y = 9,$ then the locus of the centroid of the triangle is:

(A) $x - y = 1$ (B) $2x + 3y = 1$

(C) $2x + 3y = 3$ (D) $2x - 3y = 1$

25. If $y = \sec\left(\tan^{-1}x\right),$ then $\frac{dy}{dx}$ at $x = 1$ is equal to:

(A) 1 (B) $\sqrt{2}$

(C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Space for rough work

26. The expression $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$

(A) $\tan A+\cot A$ (B) $\sec A+\text{coesc}A$

(C) $\sin A\cos A+1$ (D) $\sec A\cos ecA+1$

27. If the trivial solution is the only solution of the system of equations

$x-ky+z=0$

$kx+3y-kz=0$ then the set of all values of k is:

$3x+y-z=0$

(A) $R-\{2,-3\}$ (B) $R-\{2\}$

(C) $R-\{-3\}$ (D) $\{2,-3\}$

28. A multiple choice examination has 5 questions. Each question has three alternative answers of which exactly one is correct. The probability that a student will get 4 or more correct answers just by guessing is:

(A) $\frac{11}{3^5}$ (B) $\frac{10}{3^5}$

(C) $\frac{17}{3^5}$ (D) $\frac{13}{3^5}$

29. Consider the differential equation $y^2dx+\left(x-\frac{1}{x}\right)dy=0$. If $y(1)=1$, then x is given by:

(A) $4-\frac{2}{y}-\frac{e^{\frac{1}{y}}}{e}$ (B) $3-\frac{1}{y}+\frac{e^{\frac{1}{y}}}{e}$

(C) $1+\frac{1}{y}-\frac{e^{\frac{1}{y}}}{e}$ (D) $1-\frac{1}{y}+\frac{e^{\frac{1}{y}}}{e}$

30. $\lim\limits_{x\to 0}\frac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$ is equal to:

(A) 1 (B) 2

(C) $-\frac{1}{4}$ (D) $\frac{1}{2}$

Space for rough work

FIITJEE JEE MAINS MOCK TEST-1

Target IIT - JEE 2014

QP CODE: 120634.0

ANSWERS

SECTION – I (Physics)

1.	**B**	2.	**C**	3.	**B**	4.	**D**
5.	**A**	6.	**C**	7.	**A**	8.	**B**
9.	**B**	10.	**C**	11.	**C**	12.	**D**
13.	**D**	14.	**C**	15.	**A**	16.	**C**
17.	**B**	18.	**A**	19.	**D**	20.	**C**
21.	**B**	22.	**B**	23.	**C**	24.	**A**
25.	**B**	26.	**D**	27.	**C**	28.	**D**
29.	**A**	30.	**D**				

SECTION – II (Chemistry)

1.	**C**	2.	**D**	3.	**C**	4.	**D**
5.	**C**	6.	**B**	7.	**A**	8.	**B**
9.	**C**	10.	**B**	11.	**D**	12.	**B**
13.	**C**	14.	**A**	15.	**A**	16.	**A**
17.	**D**	18.	**C**	19.	**C**	20.	**C**
21.	**C**	22.	**C**	23.	**C**	24.	**D**
25.	**B**	26.	**B**	27.	**D**	28.	**B**
29.	**D**	30.	**A**				

SECTION – III (Mathematics)

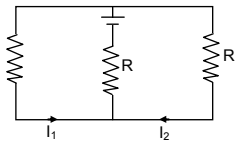
1.	**B**	2.	**D**	3.	**D**	4.	**D**
5.	**A**	6.	**C**	7.	**D**	8.	**D**
9.	**B**	10.	**A**	11.	**A**	12.	**D**
13.	**A**	14.	**A**	15.	**D**	16.	**C**
17.	**B**	18.	**A**	19.	**A**	20.	**C**
21.	**C**	22.	**C**	23.	**A**	24.	**B**
25.	**C**	26.	**D**	27.	**A**	28.	**A**
29.	**C**	30.	**B**				

Hints & Solutions

Physics

1. **B**

Sol. A moving conductor is equivalent to a battery of emf = $vB\ell$ (motion emf)



Equivalent circuit

$I = I_1 + I_2$

Applying Kirchoff's law

$I_1R + IR - vB\ell = 0$ …(1)

$I_2R + IR - vB\ell = 0$ …(2)

Adding (1) and (2)

$2IR + IR = 2vB\ell$

$$I = \frac{2vB\ell}{3R} \;;\; I_1 = I_2 = \frac{vB\ell}{3R}$$

2. **C**

Sol. $U = \frac{1}{2}\frac{q^2}{C} = \frac{1}{2C}(q_o e^{-t/T})^2 = \frac{q_0^2}{2C}e^{-2t/T}$ (where $\tau = CR$)

$$U = U_1 e^{-2t_1/\tau}$$

$$\frac{1}{2}U_1 = U_1 e^{-2t_1/\tau} \;;\; \frac{1}{2} = e^{-2t_1/\tau} \Rightarrow t_1 = \frac{T}{2}\ln 2$$

Now, $q = q_o e^{-t/T}$

$$\frac{1}{4}q_o = q_o e^{-t/2T}$$

$t_2 = T\ln 4 = 2T\ln 2$

$\therefore \quad \frac{t_1}{t_2} = \frac{1}{4}$

3. **B**

Sol. $\rho_{oil} < \rho < \rho_{water}$

Oil is the least dense of them so it should settle at the top with water at the base. Now the ball is denser than oil but less denser than water. So, it will sink through oil but will not sink in water. So it will stay at the oil-water interface.

4. **D**

Sol. $\vec{v} = Ky\hat{i} + x\hat{j}$

$\frac{dx}{dt} = Ky, \ \frac{dy}{dt} = Kx$

$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{Kx}{Ky}$

$ydy = xdx$

$y^2 = x^2 + C$

5. **A**

Sol. The magnetic field in between because of each will be in opposite direction.

$B_{\text{in between}} = \frac{\mu_o i}{2\pi x}\hat{j} - \frac{\mu_o i}{2\pi(2d-x)}(-\hat{j})$

$= \frac{\mu_o i}{2\pi}\left[\frac{1}{x} - \frac{1}{2d-x}\right](-\hat{j})$

at $x = d, \ B_{\text{in between}} = 0$

for $x < d, \ B_{\text{in between}} = (\hat{j})$

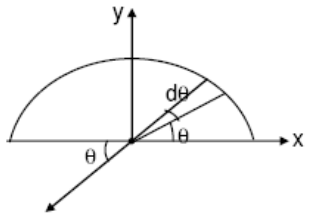
for $x > d, \ B_{\text{in between}} = (-\hat{j})$

towards x net magnetic field will add up and direction will be $(-\hat{j})$

towards x′ net magnetic field will add up and direction will be $(\hat{j})$

6. **C**

Sol. Linear charge density $\lambda = \left(\frac{q}{\pi r}\right)$



$E = \int dE\sin\theta(-\hat{j}) = \int \frac{K.dq}{r^2}\sin\theta(-\hat{j})$

$E = \frac{K}{r^2}\int \frac{qr}{\pi r}d\theta\sin\theta(-\hat{j})$

$= \frac{K}{r^2}\frac{q}{\pi}\int_0^{\pi}\sin\theta(-\hat{j}) = \frac{q}{2\pi^2\varepsilon_0 r^2}(-\hat{j})$

7. **A**

Sol. Truth table for given combination is

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	1

This comes out to be truth table of OR gate.

8. **B**

Sol. The efficiency of cycle is

$\eta = 1 - \frac{T_2}{T_1}$

for adiabatic process

$TV^{\gamma-1}$ = constant

For diatomic gas $\gamma = \frac{7}{5}$

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$T_1 = T_2\left(\frac{V_2}{V_1}\right)^{\gamma-1} \quad ; \quad T_1 = T_2(32)^{\frac{7}{2}-1}$$

$$= T_2(2^5)^{2/5} = T_2 \times 4$$

$$T_1 = 4T_2$$

$$\eta = \left(1-\frac{1}{4}\right) = \frac{3}{4} = 0.75$$

9. **B**

Sol. Apply shell theorem the total charge upto distance r can e calculated as followed

$$dq = 4\pi r^2.dr.\rho = 4\pi r^2.dr.\rho_o\left[\frac{5}{4}-\frac{r}{R}\right]$$

$$= 4\pi\rho_0\left[\frac{5}{4}r^2dr-\frac{r^3}{R}dr\right]$$

$$\int dq = q = 4\pi\rho_0\int_0^r\left(\frac{5}{4}r^2dr-\frac{r^3}{R}dr\right) = 4\pi\rho_0\left[\frac{5}{4}\frac{r^2}{3}-\frac{1}{R}\frac{r^4}{4}\right]$$

$$E = \frac{kq}{r^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{1}{r^2}\cdot 4\pi\rho_0\left[\frac{5}{4}\frac{r^2}{3}-\frac{1}{R}\frac{r^4}{4}\right]$$

$$E = \frac{\rho_o r}{4\varepsilon_o}\left(\frac{5}{3}-\frac{r}{R}\right)$$

10. **C**

Sol. $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$

$U(x=\infty) = 0$

as, $F = -\frac{dU}{dx} = -\left[\frac{12a}{x^{13}} + \frac{6b}{x^7}\right]$

at equilibrium, F = 0

$\therefore \quad x^6 = \frac{2a}{b}$

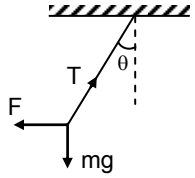
$\therefore \quad U_{\text{at equilibrium}} = \frac{a}{\left(\frac{2a}{b}\right)^2} - \frac{b}{\left(\frac{2a}{b}\right)} = \frac{-b^2}{4a}$

$\therefore \quad D = [U(x=\infty) - U_{\text{at equilibrium}}] = \frac{b^2}{4a}$

11. **C**

Sol. From F.B.D. of sphere, using Lami's theorem

$$\frac{F}{mg} = \tan\theta \qquad \ldots(i)$$



when suspended in liquid, as θ remains same,

$$\therefore \quad \frac{F'}{mg\left(1-\frac{\rho}{d}\right)} = \tan\theta \qquad \ldots(ii)$$

Using (i) and (ii)

$$\frac{F}{mg} = \frac{F'}{mgK\left(1-\frac{\rho}{d}\right)} \text{ where, } F' = \frac{F}{K}$$

$$\therefore \quad \frac{F}{mg} = \frac{F'}{mgK\left(1-\frac{\rho}{d}\right)}$$

or $$K = \frac{1}{1-\frac{\rho}{d}} = 2$$

12. **D**

Sol. Let R_0 be the initial resistance of both conductors

$\therefore$ At temperature θ their resistance will be

$R_1 = R_0(1 + \alpha_1\theta)$ and $R_2 = R_0(1 + \alpha_2\theta)$

for, series combination, $R_s = R_1 + R_2$

$R_{s0}(1+\alpha_1\theta)$ and $R_2 = (1+\alpha_1\theta) + R_0(1+\alpha_2\theta)$

where $R_{s0} = R_0 + R_0 = 2R_0$

$\therefore \quad 2R_0(1+\alpha_s\theta) = 2R_0 + R_0\theta(\alpha_1+\alpha_2)$

or $$\alpha_s = \frac{\alpha_1+\alpha_2}{2}$$

for parallel combination, $R_p = \frac{R_1R_2}{R_1+R_2}$

$$R_{p0}(1+\alpha_p\theta) = \frac{R_0(1+\alpha_1\theta)R_0(1+\alpha_2\theta)}{R_0(1+\alpha_1\theta)+R_0(1+\alpha_2\theta)}$$

where, $R_{p0} = \frac{R_0R_0}{R_0+R_0} = \frac{R_0}{2}$

$$\therefore \quad \frac{R_0}{2}(1+\alpha_p\theta) = \frac{R_0^2(1+\alpha\theta+\alpha_2\theta+\alpha_1\alpha_2\theta)}{R_0(2+\alpha_1\theta+\alpha_2\theta)}$$

as α_1 and α_2 are small quantities

$\therefore$ $\alpha_1\,\alpha_2$ is negligible

or $$\alpha_p = \frac{\alpha_1+\alpha_2}{2+(\alpha_1+\alpha)\theta} = \frac{\alpha_1+\alpha_2}{2}[1-(\alpha_1+\alpha_2)\theta]$$

as $(\alpha_1+\alpha_2)^2$ is negligible

$$\therefore \quad \alpha_p = \frac{\alpha_1+\alpha_2}{2}$$

13. **D**

Sol. $S = t^3 + 5$

$\therefore$ Speed, $v = \frac{ds}{dt} = 3t^2$

and rate of change of speed = $\frac{dv}{dt} = 6t$

$\therefore$ tangential acceleration at t = 2s, $a_t = 6 \times 2 = 12$ m/s^2

At t = 2s, $v = 3(2)^2 = 12$ m/s

$\therefore$ centripetal acceleration, $a_c = \frac{v^2}{R} = \frac{144}{20}$ m/s^2

$\therefore$ net acceleration = $\sqrt{a_t^2 + a_l^2}$

≈ 14 m/s^2

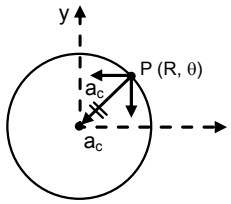
14. **C**

Sol. For a particle in uniform circular motion,

$\vec{a} = \frac{v^2}{R}$ towards centre of circle

$\therefore \quad \vec{a} = \frac{v^2}{R}(-\cos\theta\hat{i} - \sin\theta\hat{j})$

or $\quad \vec{a} = \frac{v^2}{R}\cos\theta\hat{i} - \frac{v^2}{R}\sin\theta\hat{j}$



15. **A**

Sol. 1st reaction is fusion and 4th reaction is fission.

16. **C**

Sol. Given figure is half wave rectifier.

17. **B**

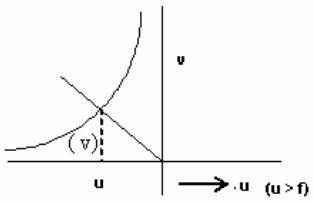
Sol. $\frac{aT}{x} = \frac{\omega^2 xT}{x} = \frac{4\pi^2}{T^2} \times T = \frac{4\pi^2}{T}$ = constant.

18. **A**

Sol. It is possible when object kept at centre of curvature.

u = v

u = 2f, v = 2f



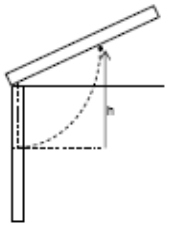
19. **D**

Sol. $T.E_i = T.E_f$

$\frac{1}{2}I\omega^2 = mgh$

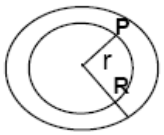
$\frac{1}{2} \times \frac{1}{3}m\ell^2\omega^2 = mgh$

$\Rightarrow \quad h = \frac{1}{6}\frac{\ell^2\omega^2}{g}$



20. **C**

Sol. $E4\pi r_1^2 = \frac{\int_0^{r_1} \frac{Q}{\pi R^4} r 4\pi r^2 dr}{\varepsilon_0} \Rightarrow E = \frac{Qr_1^2}{4\pi\varepsilon_0 R^4}$



21. **B**

Sol. Thermal energy corresponds to internal energy

Mass = 1 kg

Density = 8 kg/m^3

$\Rightarrow \quad \text{Volume} = \frac{\text{mass}}{\text{density}} = \frac{1}{8}m^3$

Pressure = 8×10^4 N/m^2

$\therefore$ Internal Energy = $\frac{5}{2}P \times V = 5 \times 10^4$ J

22. **B**

Sol. Motorcycle, u = 0, a = 2 m/s^2

Observer is in motion and source is at rest.

$$\Rightarrow \quad n' = n\frac{v - v_0}{v + v_s} \Rightarrow \frac{94}{100}n = n\frac{330 - v_0}{330} \Rightarrow 330 - v_0 = \frac{330 \times 94}{100}$$

$$\Rightarrow \quad v_0 = 330 - \frac{94 \times 33}{10} = \frac{33 \times 6}{10} m/s$$

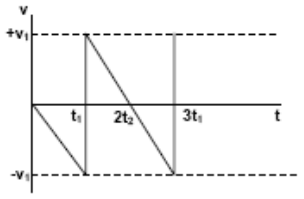
$$s = \frac{v^2 - u^2}{2a} = \frac{9 \times 33 \times 33}{100} = \frac{9 \times 1089}{100} \approx 98 \text{ m}$$

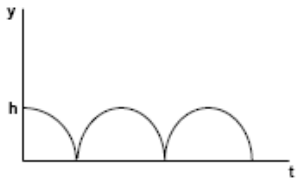
23. **C**

Sol. $h = \frac{1}{2}gt^2$, v = –gt and after the collision, v = gt

(parabolic) (straight line)

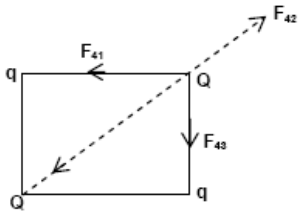
Collision is perfectly elastic then ball reaches to same height again and again with same velocity.





24. **A**

Sol. Three forces F_{41}, F_{42} and f_{43} acting on Q are shown



Resultant of $F_{41} + F_{43} = \sqrt{2}\, F_{each}$

$$= \sqrt{2}\,\frac{1}{4\pi\varepsilon_o}\frac{Qq}{d^2}$$

Resultant on Q becomes zero only when 'q' charges are of negative nature.

$$F_{4.2} = \frac{1}{4\pi\varepsilon_o}\frac{Q \times Q}{(\sqrt{2}d)^2}$$

$$\Rightarrow \quad \sqrt{2}\frac{dQ}{d^2} = \frac{Q \times Q}{2d^2}$$

$$\Rightarrow \quad \sqrt{2} \times q = \frac{Q \times Q}{2} \quad ; \quad \therefore \quad q = -\frac{Q}{2\sqrt{2}} \text{ or } \frac{Q}{q} = -2\sqrt{2}$$

25. **B**

Sol. We know that $\frac{dQ}{dt} = KA\frac{d\theta}{dx}$

In steady state flow of heat

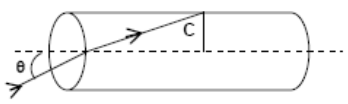
$$d\theta = \frac{dQ}{dt} \cdot \frac{1}{kA} dx$$

$\Rightarrow \quad \theta_H - \theta = k'x \quad \Rightarrow \quad \theta = \theta_H - k'x$

Equation $\theta = \theta_H - k'x$ represents a straight line.

26. **D**

Sol. $\text{Sin } C = \frac{\sqrt{3}}{2}$ …(1)



Sin r = sin (90 – C) = cos C = $\frac{1}{2}$

$$\frac{\sin\theta}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$\sin\theta = \frac{2}{\sqrt{3}} \times \frac{1}{2}$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

27. **C**

Sol. Maximum number of beats = $\nu + 1 - (\nu - 1) = 2$.

28. **D**

Sol. $A_1\ell_1 = A_2\ell_2 \Rightarrow \ell_2 = \frac{A_1\ell_1}{A_2} = \frac{A \times \ell_1}{3A} = \frac{\ell_1}{3} \Rightarrow \frac{\ell_1}{\ell_2} = 3$

$$\Delta x_1 = \frac{F_1}{A_\gamma} \times \ell_1 \qquad \ldots(i)$$

$$\Delta x_2 = \frac{F_2}{3A_\gamma}\ell_2 \qquad \ldots(ii)$$

Here $\Delta x_1 = \Delta x_2$

$$\frac{F_2}{3A\gamma}\ell_2 = \frac{F_1}{A\gamma}\ell_1$$

$$F_2 = 3F_1 \times \frac{\ell_1}{\ell_2} = 3F_1 \times 3 = 9F$$

29. **A**

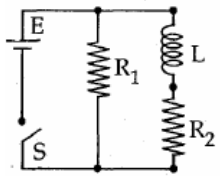
Sol. Least count = $\frac{\text{value of main scale division}}{\text{No. of divisions on vernier scale}} = \frac{1}{30}\text{MSD} = \frac{1}{30} \times \frac{1^o}{2} = \frac{1^o}{60}$ = 1 minute

30. **D**

Sol. $I_1 = \frac{F}{R_1} = \frac{12}{2} = 6A$

$$E = L\frac{dI_2}{dt} + R_2 \times I_2$$

$$I_2 = I_o(1 - e^{-t/t_c}) \Rightarrow I_o = \frac{E}{R_2} = \frac{12}{2} = 6A$$



$$t_c = \frac{L}{R} = \frac{400 \times 10^{-3}}{2} = 0.2$$

$$I_2 = 6(1 - e^{-t/0.2})$$

Potential drop across L = $E - R_2I_2 = 12 - 2 \times 6(1 - e^{-bt}) = 12\, e^{-5t}$

Chemistry

1. C

1. Ln^{3+} compounds are mostly coloured.

2. D

Sol. Esterification reaction is involved

$$CH_3COOH(\ell) + C_2H_5OH(\ell) \xrightarrow{H^+} CH_3COOC_2H_5(\ell) + H_2O(\ell)$$

3. C

Sol. Correct basic strength is $NH_3 > PH_3 > AsH_3 > BiH_3$

4. D

Sol Enthalpy of adsorption regarding physiosorption is not positive and it is negative.

5. C

Sol. $\because Fe^{3+} + 3e^- \longrightarrow Fe; E^o = -0.036\,V$

$\therefore \Delta G_1^o = -nFE^o = -3F(-0.036) = +0.108\ F$

Also $Fe^{2+} + 2e^- \longrightarrow Fe;\ E^o = -0.439\,V$

$\therefore \Delta G_2^o = -nFE^o$

= - 2F(-0.439) = 0.878F

To find E^o for $Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$

ΔG^o = -nFE^o = -1FE^o

$\because G^o = G_1^o - G_2^o$

$\therefore$ G^o = 0.108F = 0.878F

$\therefore$ -FE^o = +0.108F – 0.878F

$\therefore E^o$ = 0.878 – 0.108 = 0.77 V

6. B

Sol. $Ba(NO_3)_2 + CaCO_3 \longrightarrow BaCO_3 + 2NaNO_3$

Here $\left[CO_3^{2-}\right]$ = $[Na_2CO_3]$ = 10^{-4} M

$K_{SP} = [Ba^{2+}]\left[CO_3^{2-}\right] \Rightarrow 5.1 \times 10^{-9} = [Ba^{2+}]\,(10^{-4}) \Rightarrow [Ba^{2+}] = 5.1 \times 10^{-5}$

At this value, just precipitation starts.

7. A

Sol. Remaining are feasible

8. B

Sol. Follow the periodic trends.

9. C

Sol. Option itself is the reason

10. B

Sol. NCS^- is ambidentate ligand and it can be linked through N (or) S

11. D

Sol. $n = \dfrac{PV}{RT}$

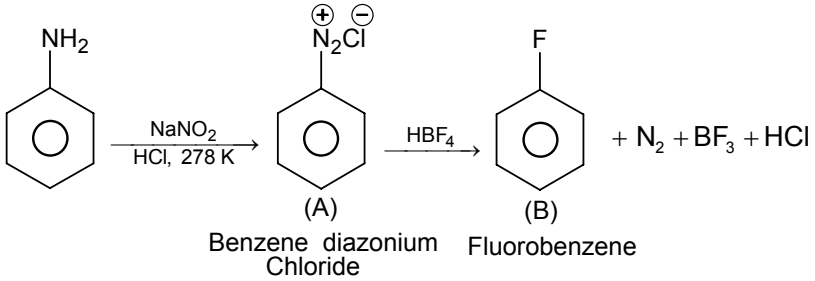
$= \dfrac{3170 \times 10^{-5}\ \text{atm} \times 1\text{L}}{0.0821\text{L atm K}^{-1}\text{mol}^{-1} \times 300\text{K}} \approx 1.27 \times 10^{-3}\ \text{mol}$

12. B

Sol. 3° alcohols react fastest with $ZnCl_2$/conc.HCl due to formation of 3° carbocation and 2–methyl propan–2–ol is the only 3° alcohol

13. C

Sol.

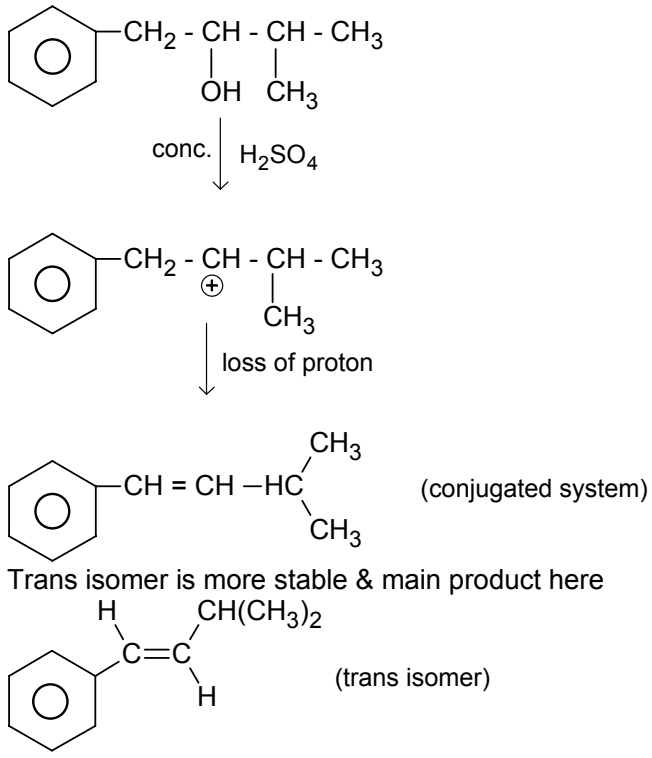


14. A

Sol. S_N1 proceeds via carbocation intermediate, the most stable one forming the product faster. Hence reactivity order for A, B, C depends on stability of carbocation

15. A

Sol.



Trans isomer is more stable & main product here

16. A

Sol. In RNA, the sugar is β - D – Ribose, whereas in DNA the sugar is β - D-2-deoxy Ribose.

17. D

Sol. $2CCl_3CHO \xrightarrow{OH^-} CCl_3COONa + CCl_3CH_2OH$

Cannizaro reaction is a disproportionation reaction. One aldehyde molecule is oxidized to salt of the carboxylic acid, other one is reduced to alcohol. So, the compound is CCl_3CH_2OH. IUPAC name is 2, 2, 2, -trichloro ethanol.

18. C

Sol. Ease of liquefication $\alpha \frac{a}{b}$

For ethane a = 5.49, b = 0.0638

For Cl_2 a = 6.49, b = 0.0562

19. C

Sol. The general o.s of lanthanides is +3, only few elements exhibit +4 o.s.

20. C

Sol. 'S' can exhibit a minimum oxidation state of -2.

21. C

Sol. In acidic medium, KBr + $KBrO_3$ in turn produces Br_2. Phenol reacts with Br_2(aq) to give 2, 4, 6-trinitrophenol.

22. C

Sol. As the size of central atom increases lone pair bond pair repulsions increases so, bond angle decreases.

23. C

Sol. $\Delta G^0 = -RT\ln K$ and $\Delta G^0 = \Delta H^0 - T\Delta S^0$

24. D

Sol. Total weight of solution = 1000 + 120 = 1120 g

$$\text{Molarity} = \frac{120}{60} \times \frac{1000}{1120/1.15} = 2.05\text{M}$$

25. B

Sol. $H_3C\text{-}CH_2\text{-}CH(CH_3)\text{-}CH_3$ on monochlorination gives

$H_2C(Cl) - CH_2 - CH(CH_3) - CH_3$
(I)
Achiral

$H_3C - CH(Cl) - CH(CH_3) - CH_3$
(II)
Chiral

$H_3C - CH_2 - \underset{\displaystyle Cl}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}} - CH_3$
(III)
Achiral

$H_3C - CH_2 - \underset{\displaystyle H}{\overset{\displaystyle CH_2 - Cl}{\overset{|}{\underset{|}{C}}}} - CH_3$
(IV)
Chiral

26. B

Sol. $$k = \frac{2.303}{40}\log\frac{0.1}{0.025}$$

$$k = \frac{0.693}{20}$$

For a 1st order reaction, rate = k(A), rate = $\frac{0.693}{20} \times 10^{-2} = 3.47 \times 10^{-4}$

27. D

Sol. FeO $\rightarrow$ More basic, more ionic, less volatile

28. B

Sol. $\Delta T_f = K_f m$

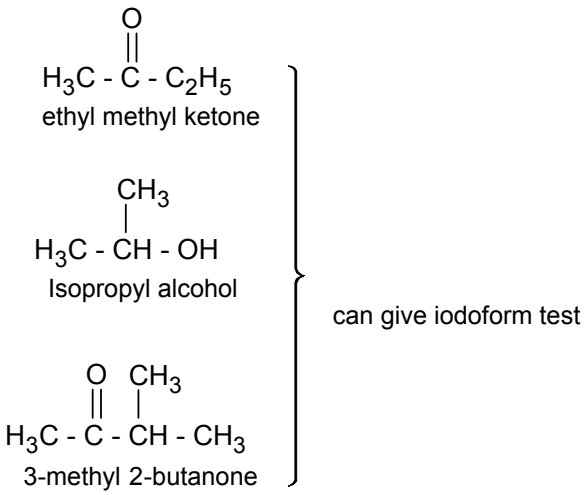
$$2.8 = 1.86 \times \frac{w}{62} \times \frac{1000}{1000}$$

w = 93g

29. D

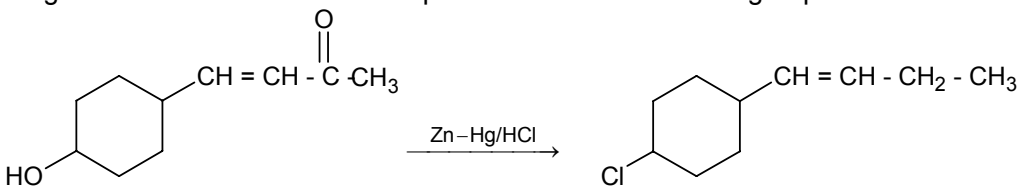
Sol. Iodoform is given by: 1) methyl ketones R – CO - CH_3
2) alcohols of the type R - $CH(OH)CH_3$

Where R can be hydrogen also



30. A

Sol. ZnHg/HCl can't be used due to the presence of acid sensitive group i.e. OH



and Na/Liq. NH_3 and $NaBH_4$ convert – CO – into – CH(OH)–

Mathematics

1. Let $f(x) = 2x^3 + 3x + k$

$f'(x) = 6x^2 + 3$

$\Rightarrow f(x)$ is an increasing function

$\Rightarrow$ there exist no real values of k

3. As we know

$|\text{adj } A| = |A|^{n-1}$ where n is order of matrix A

$$\begin{vmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{vmatrix} = (4)^2 = (12-12) - \alpha(4-6) + 3(4-6) = 16$$

$2\alpha - 6 = 16 \Rightarrow \alpha = 11$

5. $\dfrac{dp}{dx} = 100 - 12\sqrt{x} \Rightarrow \int dp = \int \left(100 - 12\sqrt{x}\right) dx + c$

$P = 100x - 8x^{3/2} + C$

At $x = 0,\ p = 2000 \quad \Rightarrow C = 2000$

$P = 100x - 8x^{3/2} + 2000$

When $x = 25$

$P = 100 \times 25 - 1000 + 2000$

$P = 3500$

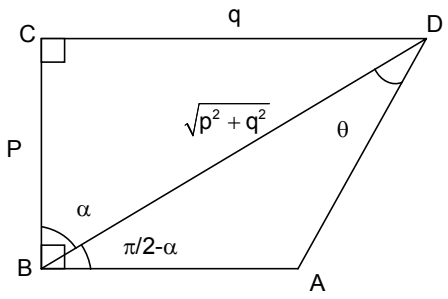
6. In ΔADB, by sine rule

$$\frac{\sin\theta}{AB} = \frac{\sin\left(\frac{\pi}{2}+\alpha-\theta\right)}{\sqrt{p^2+q^2}}$$

$$AB = \frac{\sqrt{p^2+q^2}\sin\theta}{\cos\alpha\cos\theta+\sin\alpha\sin\theta}$$

In ΔBCD, $\sin\alpha = \frac{q}{\sqrt{p^2+q^2}}, \cos\alpha = \frac{p}{\sqrt{p^2+q^2}}$

$$AB = \frac{\left(p^2+q^2\right)\sin\theta}{p\cos\theta+q\sin\theta}$$



8. Given line is $\frac{x}{\sqrt{3}}+\frac{y}{1}=1$

Passing through $\left(\sqrt{3}, 0\right)$ and $(0, 1)$, we know that image of (α,β) with respect to x axes is $(\alpha, -\beta)$

Reflected ray pass through $\left(\sqrt{3}, 0\right)$ and $(0, -1)$

$\Rightarrow \sqrt{3}y = x - \sqrt{3}$

10. Given $|z| = 1$ then

$z\bar{z} = 1$, so, $\bar{z} = \frac{1}{z}$

$$\Rightarrow \text{Arg}\ \frac{z(1+z)}{(1+z)} = \text{Arg}(z) - \theta$$

11. By Newton & libnitz rule

$f(x) = \psi'(x)$

Put $x^3 = t \Rightarrow \frac{dt}{dx} = 3x^2$

$$= \frac{1}{3}\int t\, f(t)\, dt = \frac{1}{3}\int t\,.\,\psi'(t)\,dt = \frac{1}{3}\left[t\psi(t) - \int \psi(t)\, dt\right]$$

$$= \frac{1}{3}x^3\psi\left(x^3\right) - \int x^2\psi\left(x^3\right)dx$$

13. If vector a = (1, 1, –k), b = (k, 2, 1) c = (1, –1, –1) are coplanar

then $[abc] = 0$

$$\begin{vmatrix} 1 & 1 & -k \\ k & 2 & 1 \\ 1 & -1 & -1 \end{vmatrix} = 0$$

$k^2 + 3k = 0$

15.

p	q	$\sim q$	$p \wedge \sim q$	$\sim p$	$\sim p \wedge q$	$(p \wedge nq) \wedge (\sim p \wedge q)$
T	T	F	F	F	F	F
T	F	T	T	F	F	F
F	T	F	F	T	T	F

F F T F T F F

$p \to q$	$\sim q \to \sim p$	$(p \to q) \leftrightarrow [(\sim q) \to (\sim p)]$
T	T	T
F	F	T
T	T	T
T	T	T

16. $x^2 + 2x + 3 = 0$
$D = 2^2 - 3x4 = -ve$, then both roots are common and coefficient ratio are equal
$\therefore \quad a : b : c = 1 : 2 : 3$

18. $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}}\right)^{10} = \left(\frac{\left(x^{1/3}\right)^3 + 1}{x^{2/3} - x^{1/3} + 1} - \frac{\left(x^{1/2}\right)^2 - 1}{x - x^{1/2}}\right)^{10}$

$= \left(x^{1/3} - x^{-1/2}\right)^{10}$

For term independent of x, $\frac{10-r}{3} - \frac{r}{2} = 0$

r =4

${}^{10}C_4 = 210$

20. $2\tan^{-1} y = \tan^{-1} x + \tan^{-1} z \Rightarrow \tan^{-1} \frac{2y}{1-y^2} = \tan^{-1} \frac{x+z}{1-xz}$

$y^2 = xz$

also $2y = x + z$

$\therefore x = z = y \qquad (x, y, z \text{ are in A.P. and G.P.})$

21. By Newton & Libnitz rule

$\frac{dy}{dx} = |x| = 2 \qquad \Rightarrow \qquad x = \pm 2$

At $x = 2, y = 2 \Rightarrow$ equation of tangent $y - 2 = 2(x - 2)$

At $x = -2, y = -2 \Rightarrow$ equation of tangent $y + 2 = 2(x + 2)$

So, x intercept $= \pm 1$

23. By family of point circle & line $\Rightarrow S + \lambda L = 0$

$(x-3)^2 + y^2 + \lambda y = 0$

$\lambda = -5$

$x^2 + y^2 - 6x - 5y - 9 = 0$

Which is satisfied by (5, –2)

25. By Chain rule

$y = \sec\left(\tan^{-1} x\right)$

$\frac{dy}{dx} = \sec\left(\tan^{-1} x\right).\tan\left(\tan^{-1} x\right).\frac{1}{1+x^2}$

$\frac{dy}{dx}\Big|_{x=1} = \sqrt{2} \;.\; 1.\; \frac{1}{2} = \frac{1}{\sqrt{2}}$

26. $\dfrac{\tan^2 A}{\tan A-1}+\dfrac{\cot A}{1-\tan A}\Rightarrow\dfrac{\tan^2 A-1+1}{\tan A-1}-\dfrac{\cot A}{\tan A-1}$

$=(\tan A+1)+\dfrac{1-\cot A}{\tan A-1}$

$=\tan A+1+\dfrac{\sin A-\cos A}{(\sin A-\cos A)}.\dfrac{\cos A}{\sin A}$

$=\tan A+\cot A+1$

$=\sec A.\ \mathrm{cosec}A+1$

28. By Binomial Probability

$p=\dfrac{1}{3}\quad q=\dfrac{2}{3}\quad x=5$

$p(x\geq 4)=P(x=4)+P(x=5)$

$={}^5C_4\left(\dfrac{1}{3}\right)^4\left(\dfrac{2}{3}\right)+{}^5C_5\left(\dfrac{1}{3}\right)^5$

$=\dfrac{1}{3^5}(10+1)$

$=\dfrac{11}{3^5}$

30. $\lim\limits_{x\to 0}\dfrac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$

Divide by x^2 in Numerator and denominator

$\lim\limits_{x\to 0}\dfrac{2\left(\dfrac{\sin^2 x}{x^2}\right)(3+\cos x)}{\dfrac{x\tan 4x}{x^2}}$

$\lim\limits_{x\to 0}\dfrac{2\left(\dfrac{\sin x}{x}\right)^2(3+\cos x)}{4\left(\dfrac{\tan 4x}{4x}\right)}$

$=\dfrac{2\times 1^2\times(3+1)}{4\times 1}$